Illustration 1.4

1. Let $X = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (2, 1), (2, 2), (3, 3), (1, 3), (4, 4), (1, 2), (3, 1)\}$. As $(1, 1), (2, 2), (3, 3)$ and $(4, 4)$ are all in R, it is reflexive. Also for each pair $(a, b) \in R$ the pair (b, a) is also in R. So R is symmetric. As $(2, 1), (1, 3) \in R$ and $(2, 3) \notin R$, we see that R is not transitive. Thus R is not an equivalence relation.
2. Let P denote the set of all straight lines in a plane. Let R be the relation defined on P as $\ell R m$ if ℓ is parallel to m.

   This relation is reflexive, symmetric and transitive. Thus it is an equivalence relation.
3. Let A be the set consist of parents with their children , two girls and a boy. Let R be the relation defined by aRb if a is a sister of b.

   This relation is to be looked into carefully. A woman is not a sister of herself. So it is not reflexive. It is not symmetric also. Clearly it is not transitive. So it is not an equivalence relation. (If we consider the same relation on the subset of females, then it becomes symmetric; even in this case it is not transitive).
4. On the set of natural numbers let R be the relation defined by xRy if $x + 2y = 21$. It is better to write the relation explicitly. The relation R is the set

$$\{(1, 10), (3, 9), (5, 8), (7, 7), (9, 6), (11, 5), (13, 4), (15, 3), (17, 2), (19, 1)\}.$$

   As $(1, 1) \notin R$ it is not reflexive; as $(1, 10) \in R$ and $(10, 1) \notin R$ it is not symmetric.
   As $(3, 9) \in R, (9, 6) \in R$ but $(3, 6) \notin R$, the relation is not transitive.
5. Let $X = \{1, 2, 3, 4\}$ and $R = \emptyset$, where $\emptyset$ is the empty set.

   As $(1, 1) \notin R$ it is not reflexive. As we cannot find a pair (x, y) in R such that $(y, x) \notin R$, the relation is not 'not symmetric'; so it is symmetric. Similarly it is transitive.
6. The universal relation is always an equivalence relation.
7. An empty relation can be considered as symmetric and transitive.
8. If a relation contains a single element, then the relation is transitive.

Let us discuss some more special relations now.

Example 1.10 Check the relation $R = \{(1, 1), (2, 2), (3, 3), \ldots, (n, n)\}$ defined on the set $S = \{1, 2, 3, \ldots, n\}$ for the three basic relations.

Solution:
As $(a, a) \in R$ for all $a \in S$, R is reflexive.
There is no pair (a, b) in R such that $(b, a) \notin R$. In other words, for every pair $(a, b) \in R$, (b, a) is also in R. Thus R is symmetric.
We cannot find two pairs (a, b) and (b, c) in R, such that $(a, c) \notin R$. Thus the statement "R is not transitive" is not true; therefore, the statement "R is transitive" is true; hence R is transitive.

Since R is reflexive, symmetric and transitive, this relation is an equivalence relation.

From the very beginning we have denoted all the relations by the same letter R. It is not necessary to do so. We may use the Greek letter ρ (Read as rho) to denote relations. Equivalence relations are mostly denoted by "$\sim$".

If a relation is not of required type, then by inserting or deleting some pairs we can make it of the required type. We do this in the following problem.

Example 1.11 Let $S = \{1, 2, 3\}$ and $\rho = \{(1, 1), (1, 2), (2, 2), (1, 3), (3, 1)\}$.

(i) Is ρ reflexive? If not, state the reason and write the minimum set of ordered pairs to be included to ρ so as to make it reflexive.